Computer Science & IT



Discrete Mathematics

Mathematical Logic

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Rules of inferences with respect to predicate logic
- **Topic** Important logical implications and equivalences
- **Topic** Practice questions

Slide

Topics to be Covered



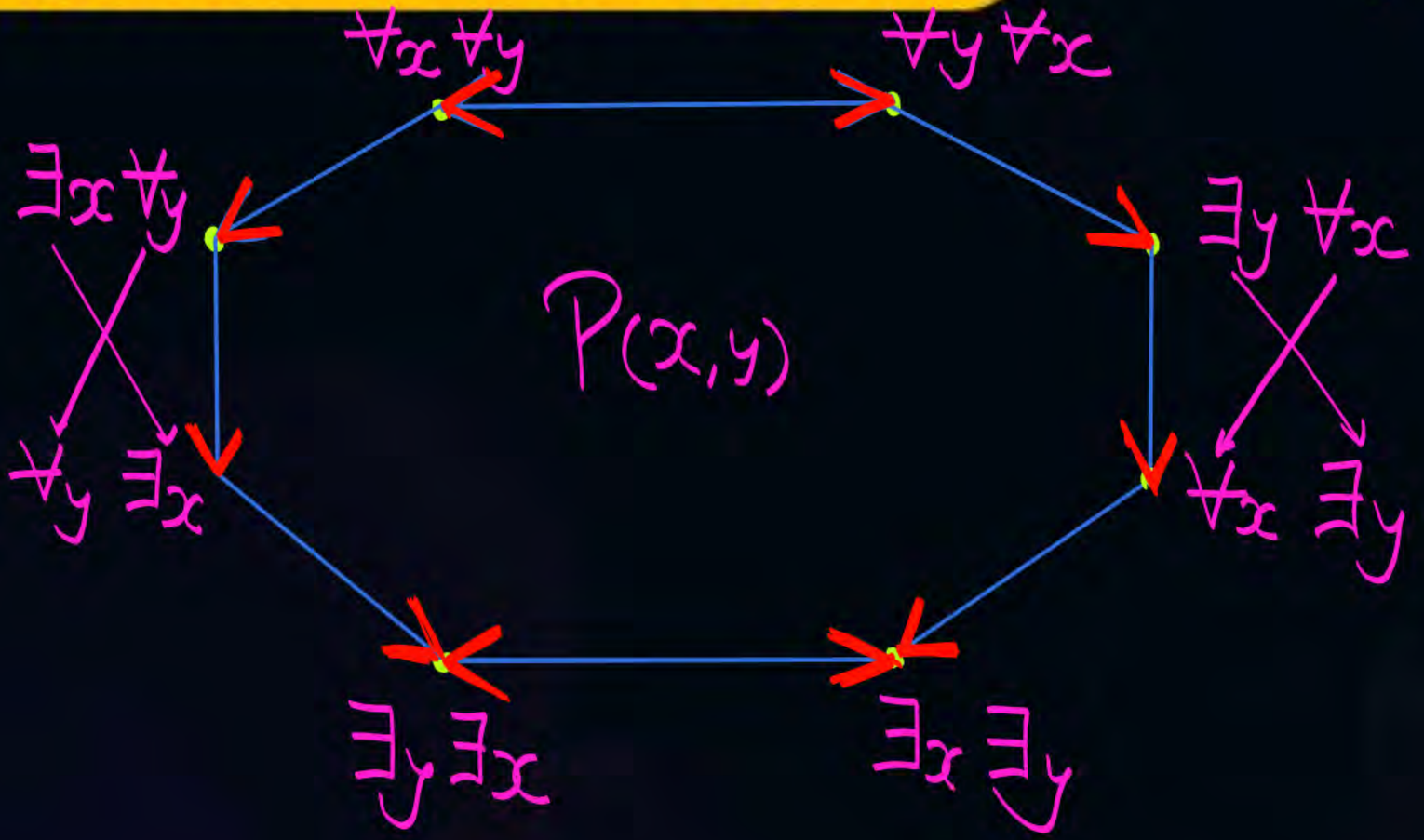
Topic — Practice questions

Topic — Tautologies w.r.t. Predicate logic

Slide



Topic : Relationship diagram



↔ represent Equivalence

&

→ represent logical implication



Topic : Rules of Inferences w.r.t. predicate logic

- In predicate logic, we can use some additional inference rules, along with all the rules of inference we have discussed in propositional logic.

- Additional Rules of Inference w.r.t. Predicate Logic

1. Universal Instantiation (Universal Specification)
2. Universal Generalization
3. Existential Instantiation (Existential Specification)
4. Existential Generalization



Topic : Universal Instantiation

Universal specification

$$\frac{\forall x \{P(x)\}}{\therefore P(a)} \text{ is valid}$$

$\therefore$ $P(a)$ is true for any 'a' in the domain

for all (or) 'a' in the domain



Topic : Universal Generalization

$P(a)$ is true for all 'a' in domain

$\therefore \forall_x \{P(x)\}$

We can use any variable for generalization

{ i.e. $\forall_z \{P(z)\}$ is also Valid }



Topic : Existential Instantiation

$$\frac{\exists_x \{P(x)\}}{\therefore P(a) \text{ is true for some 'a' in domain}} \quad \text{is Valid}$$



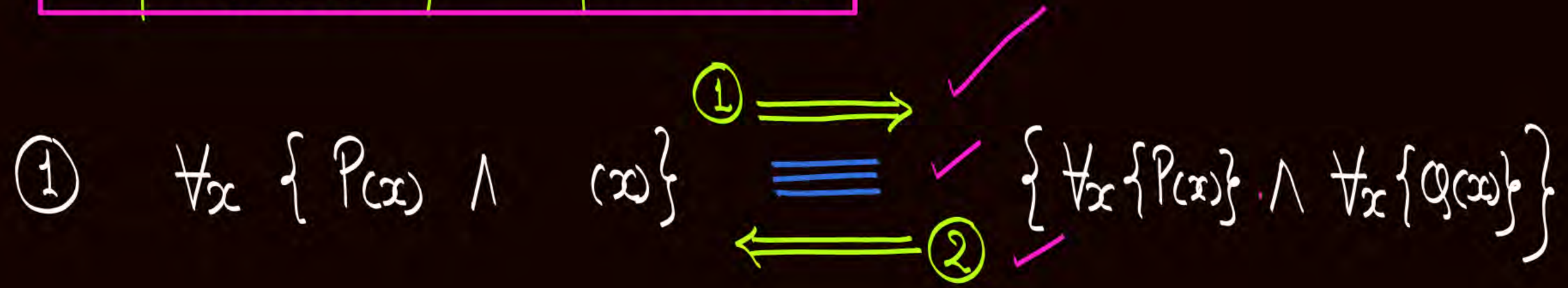
Topic : Existential Generalization

P(a) is true for Some (at least one) 'a' in domain

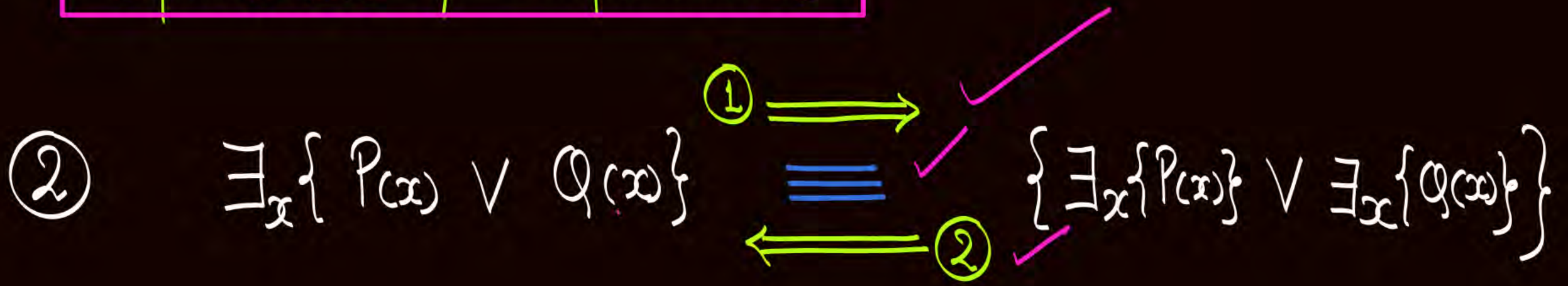
$\therefore \exists_z \{P(z)\}$

$\forall_x \{P(x)\}$ will be invalid

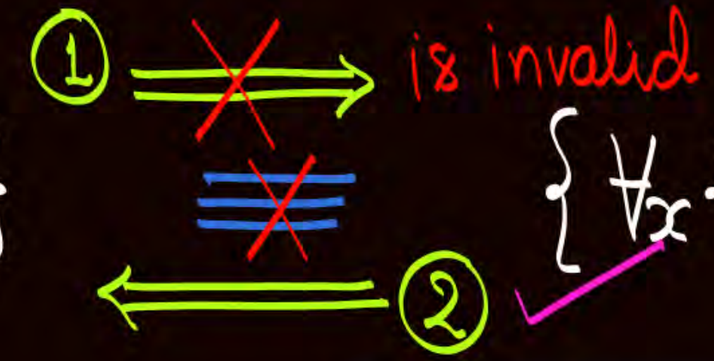
Equivalences & Implications



Equivalences & Implications



Equivalences & Implications



③ $\forall x \{ P(x) \vee Q(x) \}$ $\{ \forall x \{P(x)\} \vee \forall x \{Q(x)\} \}$

$U = \{a, b\}$

$P(a) = T \vee Q(a) = F$
$P(b) = F \vee Q(b) = T$

$T \rightarrow T$ then

LHS = True

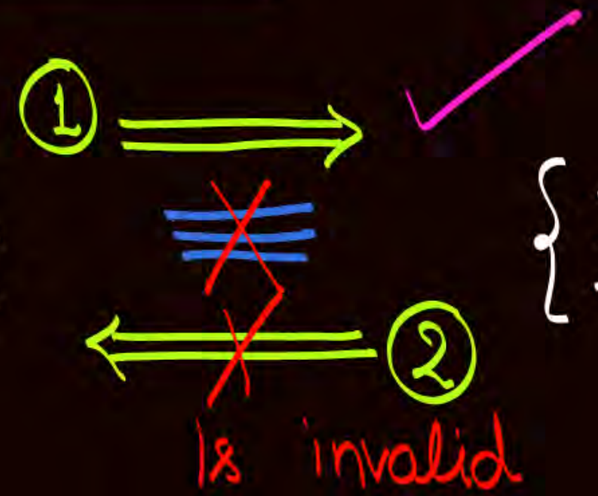
$P(a) = T$ $Q(a) = F$
$P(b) = F$ $Q(b) = T$

$\forall x \{P(x)\} = F$ $\vee$ $\forall x \{Q(x)\} = F$

$\vee$ → False

∴ RHS = False

Equivalences & Implications

④ $\exists_x \{ P(x) \wedge Q(x) \}$  $\{ \exists_x \{P(x)\} \wedge \exists_x \{Q(x)\} \}$



Topic : Equivalences & Implications

6

$\forall x \{P(x) \leftrightarrow Q(x)\} \Rightarrow (\forall x \{P(x)\} \leftrightarrow \forall x \{Q(x)\})$

1. $\forall x\,[P(x) \wedge Q(x)] \equiv [\forall x\, P(x)] \wedge [\forall x\, Q(x)]$

2. $\exists\, x\,[P(x) \vee Q(x)] \equiv [\exists\, x\, P(x)] \vee [\exists\, x\, Q(x)]$

3. $[\forall x\, P(x)] \vee [\forall x\, Q(x)] \Rightarrow \forall x\,[P(x) \vee Q(x)]$

4. $\exists\, x\,[P(x) \wedge Q(x)] \Rightarrow [\exists\, x\, P(x)] \wedge [\exists\, x\, Q(x)]$

5. $\forall x\,[P(x) \rightarrow Q(x)] \Rightarrow [\forall x\, P(x)] \rightarrow [\forall x\, Q(x)]$



Topic : Some important equivalences

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\,P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\,P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\,P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\,P(x) \wedge Q$

Note: In the given formulae 'Q' is a predicate formula without any free occurance of 'x'

⇓

i.e. Truth value of Q does not vary with 'x'

⇓

i.e. if Q is true, then it will remain true for all x

if Q is false then it will remain false for all x



Topic : Some important equivalences

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\,P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\,P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\,P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\,P(x) \wedge Q$

Case ① Q = False

When Q is false, the both LHS and R.H.S. will always return false for all values of x.

i.e. When Q = False, then same truth table for LHS & RHS

Case ② Q = True

① For LHS to be true, 'P' must be true for all x, and in that case $\forall x\{P(x)\}$ will also return true, Hence RHS will also be true

② For RHS to be true $\forall x\{P(x)\}$ must be true, Hence for all 'x' $P(x) \wedge Q$ will also be true, Hence LHS will also be true



Topic : Some important equivalences

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\,P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\,P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\,P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\,P(x) \wedge Q$

H.W. Check for validity of all the equivalences.

Hint: Check w.r.t. two cases

(i) Q = True

(ii) Q = False



Topic : Some important equivalences

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\,Q(x)$

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\,Q(x)$

In first two predicate formula 'P' is a predicate formula without any free occurance of 'x'.

7. $\forall x\,[P(x) \to Q] \equiv \exists x\,P(x) \to Q$

8. $\exists x\,[P(x) \to Q] \equiv \forall x\,P(x) \to Q$

In 7th and 8th Predicate formula 'Q' is the predicate formula without any free occurance of x.



Topic : Some important equivalences

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\, Q(x)$

$\forall x\{\sim P \vee Q(x)\} \equiv \sim P \vee \forall x\{Q(x)\} \equiv P \to \forall x\{Q(x)\} \equiv$ R.H.S.

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\, Q(x)$

$\exists x\{\sim P \vee Q(x)\} \equiv \sim P \vee \exists x\{Q(x)\} \equiv P \to \exists x\{Q(x)\} \equiv$ RHS.

7. $\forall x\,[P(x) \to Q] \equiv \exists x\, P(x) \to Q$

$\equiv \forall x\{\sim P(x) \vee Q\} \equiv \forall x\{\sim P(x)\} \vee Q$

$\sim(\exists x\{P(x)\}) \vee Q \equiv \exists x\{P(x)\} \to Q \equiv$ RHS.

8. $\exists x\,[P(x) \to Q] \equiv \forall x\, P(x) \to Q$

$\exists x\{\sim P(x) \vee Q\} \equiv \exists x\{\sim P(x)\} \vee Q \equiv \sim(\forall x(P(x))) \vee Q \equiv \forall x\{P(x)\} \to Q \equiv$ RHS

$\sim \forall x\{P(x)\}$



Topic : Some important equivalences

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\, Q(x)$

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\, Q(x)$

7. $\forall x\,[P(x) \to Q] \equiv \exists x\, P(x) \to Q$

8. $\exists x\,[P(x) \to Q] \equiv \forall x\, P(x) \to Q$

[MSQ]

#Q. Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always TRUE?

A $(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$

B $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$

C $(\forall x(P(x) \Rightarrow \forall x Q(x))) \Rightarrow (\forall x\ (P(x) \Rightarrow Q(x)))$

D $((\forall x(P(x)) \Leftrightarrow (\forall x Q(x))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x)))$

[MSQ]

#Q. Which of the following predicate calculus statements is/are valid

A $(\forall x)P(x) \vee (\forall x)Q(x) \rightarrow (\forall x)\{P(x) \vee Q(x)\}$

B $(\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)\{P(x) \wedge Q(x)\}$

C $(\exists x)\{P(x) \vee Q(x)\} \rightarrow (\forall x)P(x) \vee (\forall x)Q(x)$

(it is valid only if domain is not empty)

D $(\exists x)\{P(x) \vee Q(x)\} \rightarrow \{\exists x\, P(x) \vee \forall x\, Q(x)\}$

is invalid

if P = false for all, and Q is true for some but not for all, then LHS = True but RHS = False

[MSQ]

#Q. Which of the following predicate calculus statements is/are valid

A $\forall(x)\big(P(x) \Rightarrow Q(x)\big) \Rightarrow \big(\big(\forall(x)P(x)\big) \Rightarrow \big(\forall(x)Q(x)\big)\big)$

B $\exists x\big(P(x) \vee Q(x)\big) \Rightarrow \big(\big(\exists xP(x)\big) \Rightarrow \big(\exists xQ(x)\big)\big)$

T → F = T

if P = True for some but Q = False for all, then Lhs = True & Rhs = False

C $\exists x\big(P(x) \wedge Q(x)\big) \Leftrightarrow \big(\big(\exists xP(x)\big) \wedge \big(\exists xQ(x)\big)\big)$

D $\forall(x)\exists y\, P(x,y) \Rightarrow \exists y\, \forall(x)P(x,y)$

[MSQ]

#Q. Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

Let $\alpha \equiv P(x)$, $\beta = Q(x)$

A $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$

$\forall x\{P(x)\} \Rightarrow \forall x\{Q(x)\} \Rightarrow \forall x\{P(x) \to Q(x)\}$

B $((\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta])$

T $\to$ F $\equiv$ F

if α = True for all, but β = false for all then "$\to$" will return false

C $((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha]) \Rightarrow (\forall x)[\alpha]$

$\emptyset \to T$

T

let $\forall x(\alpha)$ = false

F = false

{ let α = false for some but not for all }

D $((\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

$\forall x\{P(x) \to Q(x)\} \Rightarrow (\forall x\{P(x)\} \to \forall x\{Q(x)\})$

Today's Topic

[MSQ]

#Q. Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and β is a first-order formula with no free variable.

A $[\beta \to (\exists x, \alpha(x))] \to [\forall x, \beta \to \alpha(x)]$

B $[\exists x, \beta \to (\alpha(x))] \to [\beta \to (\forall x, \alpha(x))]$

C $[(\exists x, \alpha(x)) \to \beta] \to [\forall x, \alpha(x) \to \beta]$

D $[(\forall x, \alpha(x)) \to \beta] \to [\forall x, \alpha(x) \to \beta]$

$\forall x\{\sim\beta \vee \alpha(x)\} \equiv \sim\beta \vee \forall x\{\alpha(x)\} \equiv \beta \to \forall x\{\alpha(x)\}$ (T)

All answers are w.r.t. non-Empty domain

(A) $\forall x\{\beta \to \alpha(x)\} \to \beta \to \exists x\{\alpha(x)\}$ (T)

(B) $\beta \to \forall x\{\alpha(x)\} \to \exists x\{\beta \to \alpha(x)\}$

$\sim\beta \vee \forall x\{\alpha(x)\}$ (F)

$\sim\beta \vee \exists x\{\alpha(x)\}$ (F)

(C) $\forall x\{\alpha(x) \to \beta\} \to \forall x\{\alpha(x)\} \to \beta$

$\forall x\{\sim\alpha(x)\} \vee \beta$

$\sim(\forall x\{\alpha(x)\}) \vee \beta$

$\exists x\{\sim\alpha(x)\} \vee \beta$

[MSQ]

H.W.

#Q. Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x.

A $\exists x(p(x) \wedge W) \equiv (\exists x\, p(x)) \wedge W$ (Valid)

B $\forall x(p(x) \rightarrow W) \equiv (\forall x\, p(x)) \rightarrow W$ (False)

C $\exists x(p(x) \rightarrow W) \equiv (\forall x\, p(x)) \rightarrow W$ (True)

D $\forall x(p(x) \vee W) \equiv (\forall x\, p(x)) \vee W$ (True)

[MSQ]

HW

#Q. What is the correct translation of the following statement into mathematical logic?

$\exists_x \{ real(x) \wedge rational(x) \}$

"some real numbers are rational"

A $\exists x\ (real(x) \vee rational(x))$

B $\exists x\ (real(x) \wedge rational(x))$

C $\forall x\ (real(x) \rightarrow rational(x))$

D $\forall x\ (rational(x) \rightarrow real(x))$

[MSQ]

#Q. What is the logical translation of the following statement?

"None of my friends are perfect"

$\forall x \{\sim F(x) \vee \sim P(x)\}$

$\equiv \sim \exists x \{F(x) \wedge P(x)\}$

Not atleast one

↳ All my friends are imperfect $= \forall x \{F(x) \rightarrow \sim P(x)\}$

A $\exists x\,(F(x) \wedge \sim P(x))$

B $\exists x\,(\sim F(x) \wedge \sim P(x))$

C $\exists x\,(\sim F(x) \wedge P(x))$

D $\sim (\exists x\,(F(x) \wedge P(x)))$

[MSQ]

H.W.

#Q. The CORRECT formula of the sentence "not all rainy days are cold"

$\sim \forall x \{ rainy(x) \rightarrow Cold(x) \}$

$\exists d \{ \sim (rainy(d) \rightarrow Cold(d)) \}$

$\exists d \{ \sim (\sim rainy(d) \vee Cold(d)) \}$

$\exists d \{ rainy(d) \wedge \sim (Cold(d)) \}$

A $\forall d\,(rainy(d) \wedge \sim cold(d))$

B $\exists d\,(\sim rainy(d) \rightarrow cold(d))$

C $\forall d\,(\sim rainy(d) \rightarrow cold(d))$

D $\exists d\,(rainy(d) \wedge \sim cold(d))$

[MSQ]

#Q. Which one the following is the most appropriate logical formula to represent the statement "gold and silver ornaments are precious". The following notations are used: $\wedge$ or $\vee$ $G(x)$:x is a gold ornament

$S(x)$:x is a silver ornament $P(x)$:x is precious

A $\forall x(P(x) \rightarrow G(x) \wedge S(x))$

B $\exists x\,((G(x) \wedge S(x)) \rightarrow P(x))$

C $\forall x\,((G(x) \wedge S(x)) \rightarrow P(x))$

D $\forall x\,((G(x) \vee S(x)) \rightarrow P(x))$

[MSQ]

H.W.

#Q. Which one of the first order predicate calculus statements given below correctly expresses the following English statements? "Tigers and lions attack if they are hungry or threatened".

All

if they or hungry or threatened then attack

A $\forall x[(tiger(x) \wedge lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$

B $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \wedge attacks(x))]$

C $\forall x[(tiger(x) \wedge lion(x)) \rightarrow (attacks(x) \rightarrow (hungry(x) \vee threatened(x)))]$

D $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$

[MSQ]

#Q. Which of the following argument is/are valid?

A $\forall x\ (P(x))$ follows from the premises $\exists x(P(x) \wedge Q(x))$

B $\exists x(P(x) \wedge Q(x))$ follows from the premises $\exists x(P(x)) \wedge \exists x(Q(x))$

C $\exists x(P(x))$ follows from the premises $\{\forall x(P(x) \Rightarrow Q(x)), \exists y(Q(y))\}$

D $\sim P(a,b)$ follows from the premises $\{\forall x\ \forall y\{\ P(x,y) \rightarrow W(x,y)\}, \sim W(a,b)\}$

(a) Existential Specification

$\exists x \{ P(x) \wedge Q(x) \}$ logically implies $\forall x \{ P(x) \}$

$\therefore$ $(P(a) \wedge Q(a))$ is true for some a

||| Simplification

$P(a)$ is true for some (a), $Q(a)$ is true for some a

$\therefore$ generalize using $\exists$ $\therefore \exists x \{P(x)\}$ is valid

$\therefore$ we can not generalize using $\forall$.

Hence $\forall x \{P(x)\}$ does not follow.

(b) $\{\exists x\{P(x)\} \wedge \exists x\{Q(x)\}\}$ logically implies $\exists x\{P(x) \wedge Q(x)\}$

We know $[\exists x\{P(x)\} \wedge \exists x\{Q(x)\}] \longrightarrow \exists x\{P(x) \wedge Q(x)\}$ is invalid

© $\{\forall x \{P(x) \rightarrow Q(x)\}, \exists y \{Q(y)\}\}$ logically implies $\exists x \{P(x)\}$

$\{\forall x \{P(x) \rightarrow Q(x)\}, \exists y \{Q(y)\}\}$ ≡ $P(a) \rightarrow Q(a)$ is true for all a

$\exists y \{Q(y)\}$ ≡ $Q(b)$ is true for some 'b'

true for all
∴ it will be true for 'b' as well

Hence $P(b) \rightarrow Q(b)$ is true

∴ $P(b)$ is true

Not valid fallacy

ⓓ $\{\forall_x \forall_y \{P(x,y) \longrightarrow W(x,y)\}, \sim W(a,b)\}$ logically implies $\sim P(a,b)$

$\Big\Vert$

$P(l,m) \longrightarrow W(l,m)$ is true for all $l \neq m$

$\downarrow$

true for all

$\therefore$ will be true for $l=a$ & $m=b$ as well

i.e. $P(a,b) \longrightarrow W(a,b)$ is true

M.T.

$\therefore P(a,b)$ is false is Valid

i.e. $\sim P(a,b)$ is true is Valid

[MSQ]

#Q. Which of the following argument is/are valid?

A $\forall x\,(P(x))$ follows from the premises $\exists x(P(x) \wedge Q(x))$

B $\exists x(P(x) \wedge Q(x))$ follows from the premises $\exists x(P(x)) \wedge \exists x(Q(x))$

C $\exists x(P(x))$ follows from the premises $\{\forall x(P(x) \Rightarrow Q(x)), \exists y(Q(y))\}$

D $\sim P(a,b)$ follows from the premises $\{\forall x\ \forall y\{P(x,y) \rightarrow W(x,y)\}, \sim W(a,b)\}$

Q:. Which of the following Predicate Formula is/are tautology

(a) $\exists x \{P(x) \wedge Q(x)\} \Longrightarrow (\exists x \{P(x)\} \wedge \exists x \{Q(x)\})$

(b) $\exists x \{P(x) \rightarrow Q(x)\} \Longrightarrow \exists x \{\sim P(x) \vee Q(x)\}$

(c) $\sim \forall x \{P(x) \wedge Q(x)\} \Longleftrightarrow \exists x \{P(x) \longrightarrow \sim Q(x)\}$

(d) $\exists x \forall y (P(x,y)) \Longrightarrow \forall y \exists x (P(x,y))$

Note:- A propositional formula which is always true is a valid propositional formula as well as a tautology.

but A predicate formula which is always true is a valid Predicate formula, but need not be a tautology.



Topic : Tautology in predicate logic

In propositional logic, tautologies and validities are same.

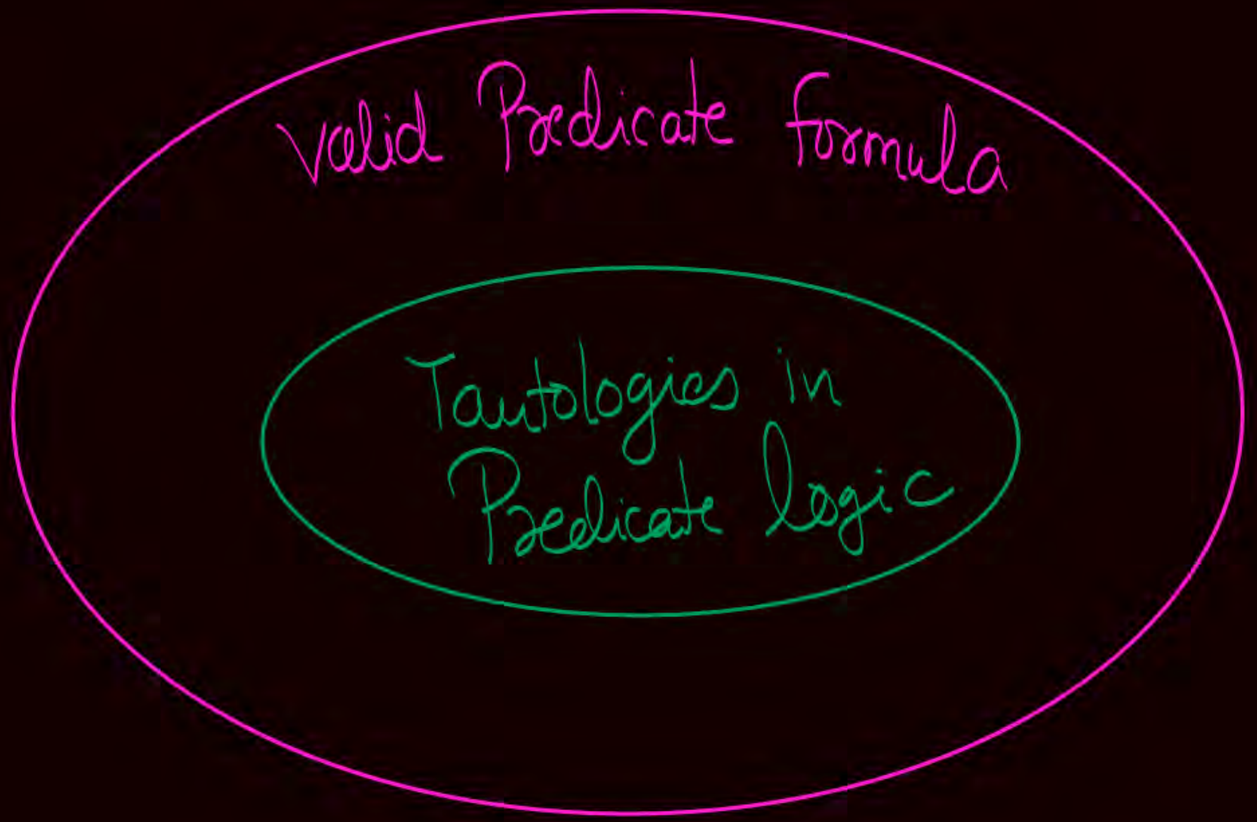
A propositional function which is always true is called valid propositional function or tautology.

But in predicate logic, distinction is maintained between logical validities and tautologies.

A predicate formula which is always true is called a valid predicate formula but it may or may not be a tautology. In predicate logic tautologies are proper subset of logical validities.

i.e.

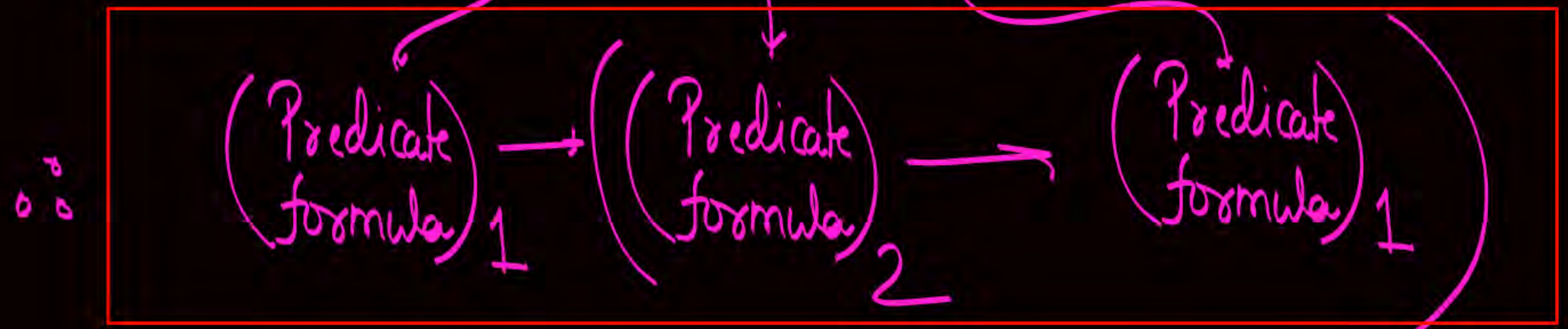




All tautologic w.r.t Predicate logic are also Valid Predicate formula but Every Valid Predicate formula need not be a tautology

If Predicate formula is invalid then it can never be a tautology

$P \rightarrow (Q \rightarrow P)$ is a tautology



is also a tautology



Topic : Tautology in predicate logic

A tautology in predicate logic is a sentence that can be obtained by taking a tautology of propositional logic and uniformly replacing each propositional variable by a predicate formula (one formula per propositional variable).

imp-

NOTE: To check whether a given predicate formula is a tautology or not we can only use the concepts that we have learned in propositional logic,

↳ i.e, We can not use any Concept from predicate logic.

Q:. Which of the following Predicate Formula is/are tautology

(a) $\exists x\{P(x) \wedge Q(x)\} \Longrightarrow (\exists x\{P(x)\} \wedge \exists x\{Q(x)\})$

P → (Q ∧ R) is not a tautology

(b) $\exists x\{P(x) \to Q(x)\} \Longrightarrow \exists x\{\sim P(x) \vee Q(x)\}$

$\exists x\{P(x) \to Q(x)\} \Longrightarrow \exists x\{P(x) \to Q(x)\}$

We know from propositional logic → P → P is a tautology

(c) $\sim \forall x\{P(x) \wedge Q(x)\} \Longleftrightarrow \exists x\{P(x) \to \sim Q(x)\}$

P ⟷ Q is not a tautology

(d) $\exists x \forall y\,(P(x,y)) \Longrightarrow \forall y \exists x\,(P(x,y))$

P → Q is not a tautology

Q:. Which of the following Predicate Formula is/are tautology

(a) $\exists x \{P(x) \wedge Q(x)\} \Longrightarrow (\exists x \{P(x)\} \wedge \exists x \{Q(x)\})$ { it is valid, but not a tautology }

(b) $\exists x \{P(x) \rightarrow Q(x)\} \Longrightarrow \exists x \{\sim P(x) \vee Q(x)\}$ { It is a tautology as well as valid }

(c) $\sim \forall x \{P(x) \wedge Q(x)\} \Longleftrightarrow \exists x \{P(x) \longrightarrow \sim Q(x)\}$ { it is valid, but not a tautology }

(d) $\exists x \forall y (P(x,y)) \Longrightarrow \forall y \exists x (P(x,y))$ { it is valid, but not a tautology }



2 mins Summary

Topic Practice questions

Topic Tautologies w.r.t. predicate logic

Slide


THANK - YOU
Slide